

NORTH SYDNEY BOYS HIGH SCHOOL

2018 HSC ASSESSMENT TASK 1

Mathematics

Extension 1

General Instructions

- Working time – 55 minutes
- Reading time – 5 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Ms Ziazaris

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	Total	Total %
Mark	$\frac{\quad}{16}$	$\frac{\quad}{11}$	$\frac{\quad}{10}$	$\frac{\quad}{9}$	$\frac{\quad}{46}$	$\frac{\quad}{100}$

(a) Find the following integrals:

(i) $\int (2x + 7)^{10} dx$ 2

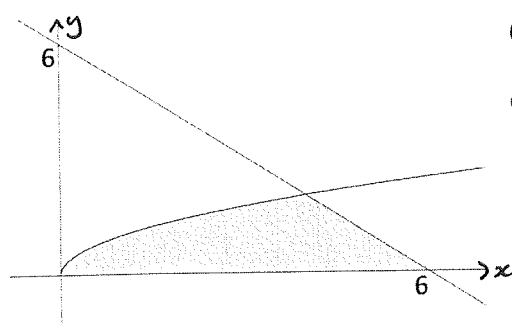
(ii) $\int \frac{5x^4 - 3}{x^2} dx$ 2

(iii) $\int x\sqrt{x} dx$ 1

(b) (i) Differentiate $(x^3 + 1)^5$ 1

(ii) Hence evaluate $\int_0^1 x^2 (x^3 + 1)^4 dx$ 2

(c)

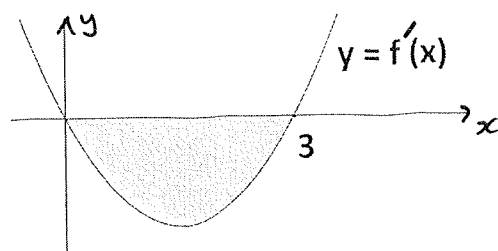


Calculate the area of the region bounded by the curves $y = \sqrt{x}$, $y = 6 - x$, and the x -axis. 3

(d) Use 5 function values and Simpson's Rule to approximate $\int_1^3 \frac{1}{1+x^2} dx$ 3

(answer to 2 decimal places)

(e)



The graph of $y = f'(x)$ is shown above, with the region bounded by $y = f'(x)$ and the x -axis shaded. Given that $y = f(x)$ has a local maximum at $(0, 7)$ and a local minimum at $(3, 4)$, find the area of the shaded region. 2

Question 2 (11 marks)

Start a NEW page

- (a) When a certain polynomial $P(x)$ is divided by $x - 3$ the remainder is 7.

When the same polynomial is divided by $x + 2$ the remainder is -3 .

What is the remainder when $P(x)$ is divided by $x^2 - x - 6$? 3

- (b) The polynomial equation $4x^3 - 12x^2 + 5x + 6 = 0$ has 3 roots, one of which is the sum of the other two. Find all three roots. 3

- (c) α, β, γ are the roots of $x^3 + 2x^2 + 3x + 4 = 0$. Evaluate the following:

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1

(iii) $\alpha\beta\gamma$ 1

(iv) $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2}$ 2

Question 3 (10 marks)

Start a NEW page

- (a) The sum of the first n terms of a series is given by the formula $\frac{n(n^2 - 1)}{3}$.

Find the formula for T_n , the n^{th} term of the series. 2

- (b) Consider the geometric series $1 + \frac{4}{3}\sin^2 x + \frac{16}{9}\sin^4 x + \frac{64}{27}\sin^6 x + \dots$

(i) When the limiting sum exists, find its value in simplest form. 2

(ii) For what values of x in the interval $0 < x < \frac{\pi}{2}$ does the limiting sum of this series exist? 2

- (c) The three positive numbers a, b , and c form an arithmetic progression, and $a + b + c = 21$.

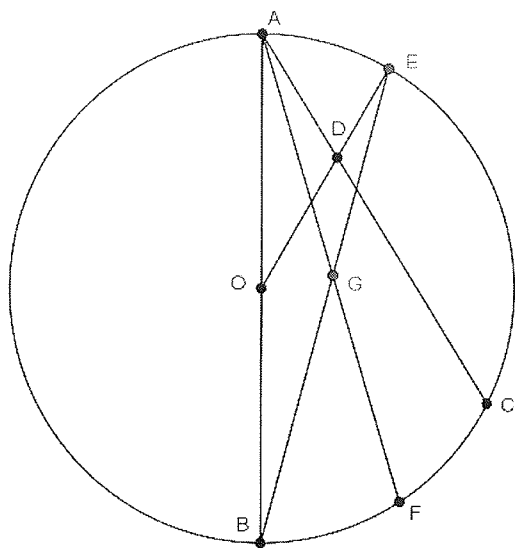
(i) Show that $b = 7$. 1

(ii) If also the numbers $a + 2, b + 3, c + 9$ form a geometric progression, then find c . 3

Question 4 (9 marks)

Start a NEW page

- (a) In the figure below, AOB is the diameter of a circle, centre O . D is a point on chord AC such that $DA = DO$, and OD is produced to E . AF is the bisector of $\angle BAC$, and cuts BE at G .



(i) Copy or trace the diagram into your answer booklet

(ii) Prove $GA = GB$ 3

(iii) Prove $AOGE$ is a cyclic quadrilateral 2

- (b) Consider the polynomial $P(x) = x^3 - kx^2 + kx - 1$, where k is a real number constant.

(i) Show that 1 is a root of $P(x) = 0$. 1

(ii) Given that α , where $\alpha \neq 1$, is a second root of $P(x) = 0$, show that

$\frac{1}{\alpha}$ is also a root of the equation. 1

(iii) Show that $\alpha^2 + \frac{1}{\alpha^2} = k^2 - 2k - 1$. 2

END OF EXAMINATION

Q1 (a) (i) $\int (2x+7)^{10} dx = \frac{(2x+7)^{11}}{2 \times 11} + C$
 $= \frac{(2x+7)^{11}}{22} + C$

✓✓
 (deduct 1 for
 no constant)

(ii) $\int \frac{5x^4 - 3}{x^2} dx = \int (5x^2 - 3x^{-2}) dx$
 $= \frac{5x^3}{3} + 3x^{-1} + C$
 $= \frac{5x^3}{3} + \frac{3}{x} + C$

✓✓

(i) $\int x \sqrt{x} dx = \int x^{\frac{3}{2}} dx = \frac{2}{5} x^{\frac{5}{2}} + C$

✓

(b) (i) $\frac{d}{dx} (x^3+1)^5 = 5 \cdot (x^3+1)^4 \cdot 3x^2$
 $= 15x^2 (x^3+1)^4$

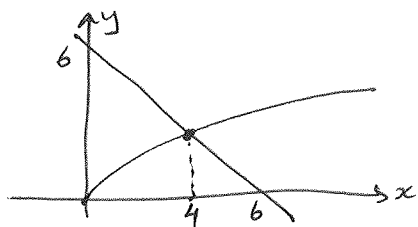
✓

(ii) $\int_0^1 x^2 (x^3+1)^4 dx = \frac{1}{15} \int_0^1 15x^2 (x^3+1)^4 dx$
 $= \frac{1}{15} \left[(x^3+1)^5 \right]_0^1$
 $= \frac{1}{15} [2^5 - 1]$
 $= \frac{31}{15}$

✓

✓

Q1 (c)



At intersection,

$$\sqrt{x} = 6 - x$$

$$x = 36 - 12x + x^2$$

$$\therefore x^2 - 13x + 36 = 0$$

$$(x - 9)(x - 4) = 0$$

$$\therefore x = 9 \text{ or } 4$$

Here, clearly $x < 6 \therefore x = 4$. (and $y = 2$)

$$\therefore \text{Area} = \int_0^4 \sqrt{x} \, dx + \frac{1}{2} \times 2 \times 2$$

$$= \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^4 + 2$$

$$= \frac{2}{3} \cdot 8 + 2 \quad \therefore \text{Area} = \frac{22}{3} \text{ units}^2$$

(d)	x	1	1.5	2	2.5	3
	$f(x)$	0.5	0.3077	0.2	0.1379	0.1

$$\int_1^3 \frac{1}{1+x^2} \, dx \doteq \frac{2-1}{6} [0.5 + 4(0.3077) + 0.2]$$

$$+ \frac{3-2}{6} [0.2 + 4(0.1379) + 0.1]$$

$$\doteq 0.46373... = 0.46 \text{ (2 d.p.)}$$

[no units]

$$(e) \text{ Area} = \left| \int_0^3 f'(x) \, dx \right| = \left| [f(x)]_0^3 \right|$$

$$= |4 - 7|$$

$$= 3 \text{ units}^2$$

Q2

$$(a) \quad x^2 - x - 6 = (x-3)(x+2)$$

$$P(x) = (x-3)(x+2) \cdot Q(x) + ax + b$$

$$P(3) = 7 \quad \therefore 3a + b = 7 \quad \text{---- (1)}$$

$$P(-2) = -3 \quad \therefore -2a + b = -3 \quad \text{---- (2)}$$

$$(1) - (2) \rightarrow 5a = 10 \quad \therefore a = 2$$

$$b = 1$$

$\therefore$ remainder is $2x + 1$.

(b) Let roots be $\alpha, \beta, \alpha + \beta$.

$$\text{Sum of roots} \rightarrow \alpha + \beta + (\alpha + \beta) = \frac{-(-12)}{4} = 3$$

$$\therefore \alpha + \beta = \frac{3}{2} \quad \text{----- (1)}$$

$$\text{Product of roots} \rightarrow \alpha \cdot \beta \cdot (\alpha + \beta) = \frac{-6}{4} = -\frac{3}{2}$$

$$\therefore \alpha\beta = -1 \quad \text{----- (2)}$$

$$\text{From (2), } \beta = -\frac{1}{\alpha}$$

$$\text{Sub. into (1)} \rightarrow \alpha - \frac{1}{\alpha} = \frac{3}{2}$$

$$\therefore 2\alpha^2 - 3\alpha - 2 = 0$$

$$(2\alpha + 1)(\alpha - 2) = 0$$

$$\therefore \alpha = -\frac{1}{2} \text{ or } 2$$

$$\beta = 2 \text{ or } -\frac{1}{2}$$

$$\alpha + \beta = 1\frac{1}{2}$$

Thus roots are $2, -\frac{1}{2}, 1\frac{1}{2}$.

Q2

(c) $x^3 + 2x^2 + 3x + 4 = 0$ has roots α, β, γ .

(i) $\alpha + \beta + \gamma = -2$

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma = 3$

(iii) $\alpha\beta\gamma = -4$

(iv) $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2} = \frac{\gamma^2 + \alpha^2 + \beta^2}{(\alpha\beta\gamma)^2}$

Now, $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$$\therefore \frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2} = \frac{(-2)^2 - 2(3)}{(-4)^2}$$

$$= \frac{4-6}{16}$$

$$= -\frac{1}{8}$$

Q3

(a) $T_n = S_n - S_{n-1}$

$$= \frac{n(n^2-1)}{3} - \frac{(n-1)[(n-1)^2-1]}{3}$$

$$= \frac{n(n^2-1)}{3} - \frac{(n-1)(n^2-2n+1-1)}{3}$$

$$= \frac{n^3-n}{3} - \frac{(n-1)(n^2-2n)}{3}$$

$$= \frac{\cancel{n^3}-n - \cancel{n^3} + n^2 + 2n^2 - 2n}{3}$$

$$= \frac{3n^2-3n}{3}$$

$$\therefore T_n = n^2 - n.$$

Q3 G.P. is $1 + \frac{4}{3} \sin^2 x + \frac{16}{9} \sin^4 x + \dots$

(b) (i) $S_{\infty} = \frac{a}{1-r} = \frac{1}{1 - \frac{4}{3} \sin^2 x}$

$\therefore S_{\infty} = \frac{3}{3 - 4 \sin^2 x}$

(ii) S_{∞} exists when $|r| < 1$

$\therefore \left| \frac{4}{3} \sin^2 x \right| < 1$

$\therefore \sin^2 x < \frac{3}{4} \quad \therefore 0 < \sin x < \frac{\sqrt{3}}{2}$

$\therefore 0 < x < \frac{\pi}{3}$

(c) a, b, c is an A.P., and $a+b+c=21$

(i) rewrite the AP as $b-d, b, b+d$

Then $(b-d) + b + (b+d) = 21$
 $3b = 21$

$\therefore b = 7.$

(ii) The GP is $a+2, 10, c+9$ (as $b=7$)

$\therefore \frac{10}{a+2} = \frac{c+9}{10} \quad \therefore (a+2)(c+9) = 100$

But $c-7 = 7-a \quad \therefore c = 14-a.$

Subbing in, $(a+2)(23-a) = 100$

$\therefore 23a - 2a + 46 - a^2 = 100$

$\therefore a^2 - 21a + 54 = 0$

$(a-18)(a-3) = 0 \quad \therefore a = 18 \text{ or } a = 3.$

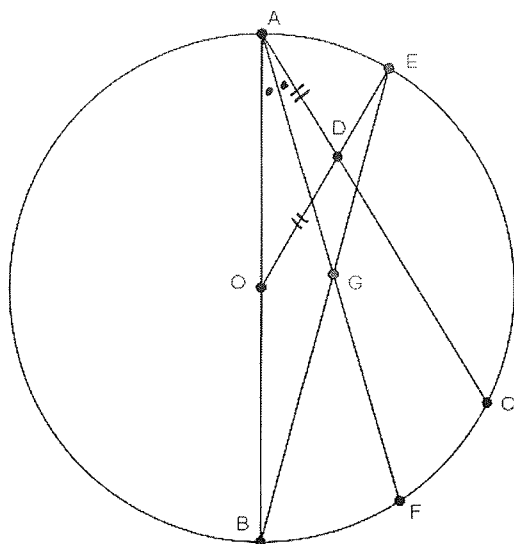
So the AP could be $3, 7, 11$ or $18, 7, -4.$

But all numbers $> 0 \quad \therefore$ series is $3, 7, 11$

$\therefore c = 11.$

Q4

(i)

(ii) Let $\angle GBA = x$

$\therefore \angle AOE = 2x$ [angle at centre is twice the angle at the circumference standing on the same arc]

But $\angle AOE = \angle OAD$ [base angles of isosceles $\triangle ADO$]

$= 2 \times \angle OAF$ [given]

$\therefore \angle OAF = x = \angle GBA$

$\therefore GA = GB$ [equal sides opposite equal angles in $\triangle AGB$]#

(iii) $OE = OB$ [equal radii]

$\therefore \angle OEB = \angle OBE$ [base angles of isosceles $\triangle OBE$]
 $= x$ (from part (i))

$\therefore \angle OEG = \angle OAG$

$\therefore A O G E$ is cyclic [2 points A & E on same side of interval OG subtend equal angles.]#

Q4

(b) $P(x) = x^3 - kx^2 + kx - 1$

(i) $P(1) = 1^3 - k \cdot 1^2 + k \cdot 1 - 1$

$$= 1 - k + k - 1$$

$$= 0$$

$\therefore x=1$ is a root of $P(x)=0$.

✓

(ii) α is a root $\Rightarrow \alpha^3 - k\alpha^2 + k\alpha - 1 = 0$

Now, $P\left(\frac{1}{\alpha}\right) = \left(\frac{1}{\alpha}\right)^3 - k\left(\frac{1}{\alpha}\right)^2 + k\left(\frac{1}{\alpha}\right) - 1$

$$= \frac{1}{\alpha^3} - \frac{k}{\alpha^2} + \frac{k}{\alpha} - 1$$

Now $\alpha \neq 0$ (since $P(0) \neq 0$), so multiply both sides by α^3 :

$$\alpha^3 \cdot P\left(\frac{1}{\alpha}\right) = 1 - k\alpha + k\alpha^2 - \alpha^3$$

$$= -1(\alpha^3 - k\alpha^2 + k\alpha - 1)$$

$$= -1 \times 0$$

$$= 0$$

since $P(\alpha) = 0$.

$\therefore$ Since $\alpha^3 \neq 0$, $P\left(\frac{1}{\alpha}\right) = 0 \quad \therefore \frac{1}{\alpha}$ is a root.

✓

(iii) Sum of roots is $\alpha + \frac{1}{\alpha} + 1 = k$

✓

$$\therefore \alpha + \frac{1}{\alpha} = k - 1$$

$$\therefore \left(\alpha + \frac{1}{\alpha}\right)^2 = (k-1)^2$$

$$\alpha^2 + \frac{1}{\alpha^2} + 2 = k^2 - 2k + 1$$

$$\therefore \alpha^2 + \frac{1}{\alpha^2} = k^2 - 2k - 1$$

✓